Week 2

Dataset Preparation and Baseline Classifier

1 Environment and Files

Week 2 is split into two notebooks. Notebook A (`Week2_A_DataPrep.ipynb`) combines the Week 1 output with two new files into one prepared dataset. Notebook B (`Week2_B_Baseline.ipynb`) turns every review into a vector with a pretrained multilingual encoder, trains two simple classifiers on those vectors and scores them. The runs used Python 3.9.6, pandas 2.3.3, numpy 2.0.2, scikit-learn 1.6.1, PyTorch 2.8.0 and transformers 4.57.6 (listed in `requirements.txt`). The encoder ran on the Apple `mps` device, and the random seed is 42. Notebook B also writes `run_settings.json`, which records the settings, the chosen values, the library versions and the SHA-256 hash of the dataset file,

`d919c0208d1b9ad9d22f1401756e1b9985ba8bdd3a1e134a7c373cac04e5c632`

so that every score can be tied to the exact file that produced it. The Samanantar sample from Week 1 is not used in either notebook.

2 Inputs

Notebook A reads three files and stops with an error if a required column is missing:

- the processed sentiment file from Week 1 (3,468 rows; columns `language`, `split`, `text`, `label`);
- `split_manifest.csv` (1,152 rows), which gives each review an identifier, its English text, its sentiment label and class number, and the part (`train`, `val` or `test`) it belongs to;
- `intent_labels.csv` (1,152 rows), which gives each review an intent label and an `unsure` flag, together with two columns that record a second review pass (`reviewed_pass2`, `changed_in_pass2`).

3 Linking the Translations

Week 1 ended with a problem: the three languages contain translations of the same reviews, but the saved file has no key that links them. Notebook A builds that key. The identifier of a review is its original split followed by its position inside its language

and split, for example `validation_0` or `test_936`. Because the identifier comes from row order, it is only valid if the three languages are aligned row by row, so the notebook checks this in several ways. Hindi, Bengali and Marathi must have exactly the same set of identifiers. The sentiment label of every review must be the same in all three languages (a label missing in only one language also counts as a difference). The identifiers, labels and original split in the manifest must agree with Week 1. Every identifier must be unique in the manifest and in the intent file, and the two files must contain the same identifiers.

A further check compares the rows with the English reviews in the manifest. The numbers that occur in an English review should also occur in its translation. For the same row this holds for 98% of the reviews in Hindi, 97% in Bengali and 96% in Marathi, while for the neighbouring row it holds for only about 1%. The notebook stops if the same-row share falls below 90% or the neighbouring-row share rises above 20%. This is indirect evidence, because only reviews that contain a number can be tested, but a shifted alignment would be detected.

4 Reviews Left Out

Week 1 has 1,156 reviews per language, and the dataset has 1,152. The four missing reviews are all from the original test split, and the notebook requires a stated reason for each one:

- `test_663` and `test_694` have no sentiment label;
- `test_662` and `test_744` repeat another review (the text is identical to a kept review in all three languages).

These are the same cases that Week 1 reported as six missing labels and six duplicated texts, counted per language rather than per review. If a review were absent from the manifest without being unlabelled or repeated, the notebook would raise an error. The dataset therefore has

$$N = 3 \times 1152 = 3456$$

rows, which is 1,152 reviews in each of the three languages.

5 Class Labels

Sentiment. The manifest gives Negative the class number 0 and Positive the class number 1. As in Week 1, no Neutral label exists, so sentiment is a two-class problem and the three-class head planned for this stage is not possible with these data.

Intent. The allowed intents are Inquiry, Opinion and Factual, and the notebook checks that every review has exactly one of them. Only two occur: 1,057 reviews (91.8%) are Opinion and 95 (8.2%) are Factual. There is no Inquiry review. A class without examples gets no class number, so Opinion is 0 and Factual is 1. The intent labels are silver labels and not human-verified ground truth, and the notebook prints this note when it saves the dataset. The `unsure` flag is set for 133 reviews (11.5%), and it is far more common

among Factual reviews (55 of 95, 58%) than among Opinion reviews (78 of 1,057, 7%). A second pass looked at 354 reviews and changed the intent of 15 of them.

6 Splitting by Review

The split is made per review, as Week 1 recommended, and every review keeps its part in all three languages. Each language therefore has 806 training, 173 validation and 173 test reviews (70%, 15% and 15%). The proportions of the labels are almost the same in the three parts (Table 1).

	Train	Val	Test	Total
Reviews per language	806	173	173	1,152
Negative	400	86	86	572
Positive	406	87	87	580
Opinion	740	157	160	1,057
Factual	66	16	13	95

Table 1: Reviews per part. The counts are the same in Hindi, Bengali and Marathi.

The new parts do not follow the original Week 1 splits. Of the 156 reviews from the original validation split, 106 are in train, 28 in val and 22 in test; of the 996 kept reviews from the original test split, 700 are in train, 145 in val and 151 in test. Notebook A checks that no review is in more than one part, that no text appears in more than one part within a language, that each part contains the same reviews in all three languages, and that the sentiment label of every row equals its Week 1 label. The prepared data are saved as `week2_dataset.csv` with the columns `review_id`, `language`, `source_split`, `text`, `part`, `label`, `label_id`, `intent`, `intent_id` and `unsure`.

7 Baseline Model

The baseline is deliberately simple. The pretrained encoder `xlm-roberta-base` is used as it is and is not trained. Each cleaned review is split into subword tokens by the encoder's own tokenizer and cut after 128 tokens. The encoder returns a vector for every token, and the vectors of the real tokens (not the padding) are averaged into one 768-number vector per review. These vectors are saved in `vectors_xlm-roberta-base.npz` together with a hash of the input texts, so that they are reused only when the model, the length limit and the texts are unchanged. The rule-based tokens from Week 1 are not used by this model.

For each task (sentiment and intent) a standardisation step followed by a logistic regression is fitted. Only the 806 Hindi training reviews are used for fitting. The regularisation setting C is chosen from $\{0.01, 0.1, 1, 10\}$ by the macro-F1 on the 173 Hindi validation reviews, and the smaller C is kept on a tie (Table 2). Bengali and Marathi are never used for training or for choosing C, so their scores measure transfer from Hindi to languages

the classifier has not seen. A majority-class model, which ignores the text and always gives the most common training class, is scored as a reference.

Validation macro-F1	$C = 0.01$	$C = 0.1$	$C = 1$	$C = 10$
Sentiment	**0.925**	0.902	0.896	0.884
Intent	0.660	**0.687**	0.677	0.626

Table 2: Choice of C on the Hindi validation reviews. The chosen value is in bold.

Every model is scored on the validation and the test reviews of each language, once on all reviews and once on the reviews that are not flagged `unsure`. The scores are accuracy, macro-F1 (the average F1 over the classes that have examples) and the precision, recall and F1 of each class. They are saved as `metrics.csv` and `per_class_metrics.csv`. Test reviews are not used for any choice.

Relation to the project plan. The plan for Week 2 names XLM-RoBERTa as the model, which the baseline follows. It also describes a single shared encoder with two classification heads, one for sentiment (Positive, Negative, Neutral) and one for intent (Inquiry, Opinion, Factual), together with mixed-precision training and gradient accumulation. The baseline differs in three ways. The encoder is shared in the sense that both tasks use the same stored vectors, but it is frozen, and the two classifiers are fitted separately rather than trained together. Mixed precision and gradient accumulation were not needed, because the encoder is only run forward once to compute the vectors and no encoder weights are updated. The sentiment head has two classes and the intent head has two classes, because Neutral and Inquiry have no examples. Joint training of the encoder with both heads is therefore left for the fine-tuning stage.

8 Results

Table 3 gives the scores on the 173 test reviews of each language. Hindi is the language the classifiers were trained on.

		Macro-F1			Accuracy		
Task	Model	hi	bn	mr	hi	bn	mr
Sentiment	Majority class	0.335	0.335	0.335	0.503	0.503	0.503
Sentiment	XLM-R + logistic	**0.942**	**0.878**	**0.884**	0.942	0.879	0.884
Intent	Majority class	0.480	0.480	0.480	0.925	0.925	0.925
Intent	XLM-R + logistic	**0.697**	**0.627**	**0.536**	0.925	0.803	0.913

Table 3: Test scores on all 173 reviews per language.

Sentiment. The classifier is far above the majority-class reference in every language. It reaches a macro-F1 of 0.942 in Hindi and about 0.88 in Bengali and Marathi, so roughly six points are lost in the languages it never saw. With only 173 test reviews this

difference is uncertain. Intervals from resampling the test reviews 2,000 times (seed 42) are 0.905–0.976 for Hindi, 0.823–0.925 for Bengali and 0.838–0.925 for Marathi, and they overlap. These intervals are not computed in the notebooks; they were obtained from the saved vectors. In Bengali the errors lean one way: 100 reviews are predicted Positive and 73 Negative, against 87 and 86 in the data.

Intent. The macro-F1 is above the majority-class value of 0.480 in all three languages, but the accuracy is not. Because 92.5% of the test reviews are Opinion, always answering Opinion is already 92.5% accurate; the classifier matches this in Hindi and is lower in Bengali (0.803) and Marathi (0.913). The result depends almost entirely on the 13 Factual test reviews (Table 4). In Hindi the classifier finds 5 of them while predicting Factual 10 times. In Bengali it finds 10 but predicts Factual 41 times, so most of these predictions are wrong. In Marathi it finds 1 of 13 and predicts Factual only 4 times. The macro-F1 intervals are wide (Hindi 0.553–0.824, Bengali 0.532–0.715, Marathi 0.468–0.660).

Factual class	Precision	Recall	F1	Support	Predicted Factual
Hindi	0.500	0.385	0.435	13	10
Bengali	0.244	0.769	0.370	13	41
Marathi	0.250	0.077	0.118	13	4

Table 4: Per-class test scores for the Factual intent.

Reviews not flagged unsure. Leaving out the 25 test reviews flagged `unsure` gives 148 reviews. The sentiment macro-F1 hardly changes (0.952, 0.872 and 0.885). For intent the macro-F1 is 0.615, 0.580 and 0.490, against 0.493 for the majority class. Nine of the 13 Factual test reviews are flagged `unsure`, so only 4 Factual reviews remain in this subset, and the intent scores there carry very little information.

9 Findings and Next Steps

Notebook A produces one dataset in which every review has a key, one sentiment label, one intent label and a fixed part, identical in the three languages. Notebook B shows that a frozen multilingual encoder with a linear classifier, trained on Hindi alone, predicts sentiment well in all three languages. It does not yet predict intent reliably. The main limitations, and the work they imply, are as follows.

- **Two sentiment classes only.** There is no Neutral class in the data, so a three-class sentiment head still needs another source or a change of design.
- **Intent labels are silver and one class is rare.** Inquiry never occurs and Factual makes up 8.2% of the reviews, with only 13 in the test part of each language. The Factual scores can change a lot with a few reviews, and a check of the Factual labels by hand, or more Factual examples, is needed before the intent result is trusted.
- **Hindi is the only training language.** Whether training on Bengali and Marathi reviews as well would reduce the loss in those languages has not been tested, although the notebook has a setting for it. The encoder is also not fine-tuned. The plan for Week 3 unfreezes the encoder and fine-tunes it, trains mainly on Hindi and tests transfer to Bengali and Marathi, and adds probability and confidence scores;

the scores in Table 3 are the reference that fine-tuning has to improve on. Accuracy is reported here, and for single-label classification it equals micro-F1, which the plan also asks for.

- **Edge of the grid.** The chosen $C = 0.01$ for sentiment is the smallest value tried, so a smaller value was not examined.